



DEALER BEST PRACTICE SERIES

Exhaust Manifold Lifting  
Tool - C175 Engines

| Maintenance and<br>Repair | Application | Component Life<br>Management | Component<br>Renewal (CRC) | MARC<br>Management                                 |
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**Editor’s Note:** The use of this installation tool and the processes discussed within this Best Practice is not a substitute for applicable laws, standards and regulations and does not alter or limit the obligation of individuals who utilize the tool to fully comply with federal, state and local law and prudent safety measures relating to the use of lifting devices.

## 1. Introduction

Ferreyros has developed a simple tool to improve the process of assembling the C175 engines in installing the exhaust manifold. The tool allows greater stability during installation and prevents the exhaust manifold to flex during lifting. Align the contact faces exhaust manifold with respect to the cylinder head, providing the proper angle, reducing the time and effort applied by the technician. Caterpillar does not have a special tool for this process.

## 2. Best Practice Description

The following photos show the fixture and process.



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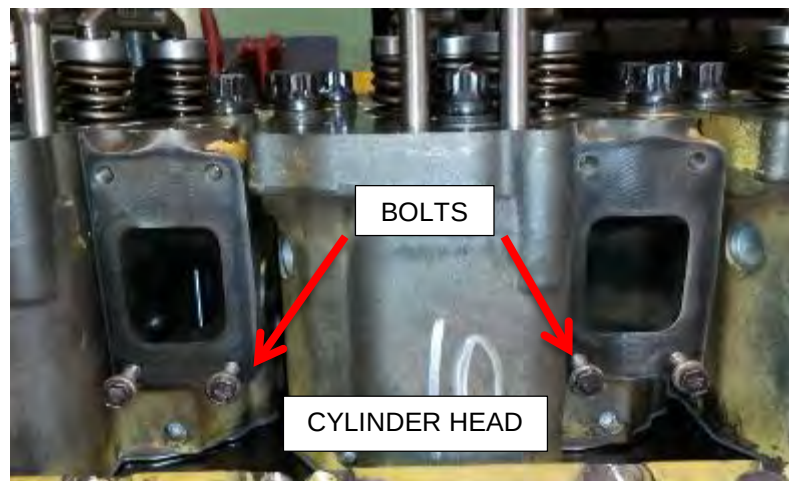
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The technician proceeds to install the threads of the bolts to secure the exhaust manifold cylinder head in the lower part.



Proceed to install the tool on the exhaust manifold and lift the component with a sling to align the contact faces exhaust manifold with respect to the cylinder head.



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Install the gasket and the bolts securing the exhaust manifold to the cylinder head at the top. Remove the tool and apply the appropriate torque to the bolts securing the exhaust manifold.



The exhaust manifold is safely installed without losing its stability at all times. This same tool should be used for the removal process, because it minimizes risks during the operation for the exhaust manifold and the people around.

### 3. Implementation Steps

Design the tool.

### 4. Benefits

- Improve operator safety.
- Improve process efficiency.
- Prevents damage to the exhaust manifold during remove or install.

### 5. Resources Required

- Minimal expense to fabricate the tool.

### 6. Supporting Attachments / References

Prints

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## 7. Related Best Practices

None

## 8. Acknowledgements

This Best Practice was written by:

Jhon Meneses Cuzcano  
Ferreyros S.A.

The idea for this Best Practice belongs to:

David Huamanchumo Urcia  
Pomahuacre Ramírez Eder  
Mateo Bernardo John  
Ferreyros S.A.

This Best Practice was revised by:

Aninu Mitacc Vizcardo  
Ferreyros S.A.

[aninu.mitacc@ferreyros.com.pe](mailto:aninu.mitacc@ferreyros.com.pe)

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